CO-3 Home Assignment - 03

① Inspect the design of a pedestrain detection sys-tem for an autonomous vehicle using convolutional neutral network models & real-world street video data containing occlusions, motion blur, & varying illum-ination, & analyze how convolution layers, pooling strateg-ies, & feature hierarchies influence detection reliability & system response time.

(6M)

6

Block Diagram :-

Street vedio Input
↓
Frame extraction & Preprocessing.
↓
Conolution Layers
↓
Pooling Layers → Feature Hierarchy Learning
↑
Fully connected / Detection Layer
↑
Pedestrain Detection output.

⊛ convolution Layers

Extract important visual features such as edges, textures and human body shape, helping detect pedestrains even with motion blur and illumination variations.

⊛ Pooling stratergies

max/average pooling reduces feature map size, improving computational efficiency and response time while preserving important features.

⊛ Feature Hierarchies

CNN learns multi-level features:

Early layers → edges and corners

Middle layers → body parts

Deeper layers → full pedestrain structure.

⊛ Handling Real-world challenges

Deep CNN models trained on street datasets improve robustness against occlusions, lighting changes, & background clutter

⊛ Detection Reliability v/s Response time

Deeper networks improve accuracy, but Tes latency. optimized architectures balance real-time detection and reliability.

(2) (6M) Distinguish the performance charcteristics of a deep learning-based road scene segmentation pipeline using camera input from urban driving environments by demonstrating how the CNN architecture segments lanes, vehicles, and obstacles, and evaluating system performance under challenging conditions such as rain, night driving, and dense traffic.

1. Camera Input & Preprocessing.

+ The system captures road images from a front-mounted camera in urban environments.

+ Images are resized, normalized, and enhanced before feeding into the CNN model.

2. CNN Feature Extraction.

+ Convolutional layers detect spatial features such as lane markings, vehicles, pedestrians, and obstacles.

+ Pooling layers reduce dimensionality while preserving important patterns.

Segmantic Segmentation Process

+ Encoder-decoder CNN architecture (e.g., U-Net/ segNet) classifies each pixel into categories.

+ output map labels regions like lanes, vehicles, road surface, and obstacles.

Real-Time Detection Performance

+ The system must process frames quickly (≈ 20-30 FPS) for autonomous driving applications.

+ GPU acceleration improves inference speed & accuracy.

Performance in challenging Conditions.

+ Rain: water droplets & reflections may reduce lane visibility.

+ Night driving: Low illumination affects camera clarity and feature detection.

+ Dense traffic: Overlapping objects increase segm-entation complexity.

Evaluation Metrics

+ Performance is measured using IOU (Intersection over Union), pixel accuracy, and latency.

+ Robust models maintain high segmentation ~~complexity~~ accuracy even under adverse conditions.

Block Diagram

camera Input
↓
Image Preprocess
↓
CNN Feature Extraction
↓
Semantic Segmentation CNN → segmented output Map

Tests (Rain/ Night / Traffic)
IoU, Accuracy, FPS
Performance Evaluation

(3) (6M) Examine the construction of a real-time semantic scene segmentation system using convolutional neural networks to interpret complex road environments containing pedestrians, vehicles, & dynamic obstacles, & illustrate how segmentation outputs support decision-making in autonomus driving systems.

Real-Time Semantic Scene Segmentation using CNN for Autonomous Driving.

1. Image Acquistion

A front-facing camera captures real-time road images from the driving environment containing pedestrains, vehicles, lanes, and obstacles.

2. Preprocessing stage

Images are resized, normalized, and enhanced to improve clarity before being processed by the neural network.

3. CNN Feature Extraction

Convolutional Neural Networks extract important spatial features such as edges, shapes, pedestrains, vehicles, & road boundaries using convolution & pooling layers.

4. Semantic Segmentation

An encoder-decoder CNN architecture assigns a class lable to every pixel in the image, producing a segmented map of road, vehicles, pedestrains, traffic signs, and obstacles.

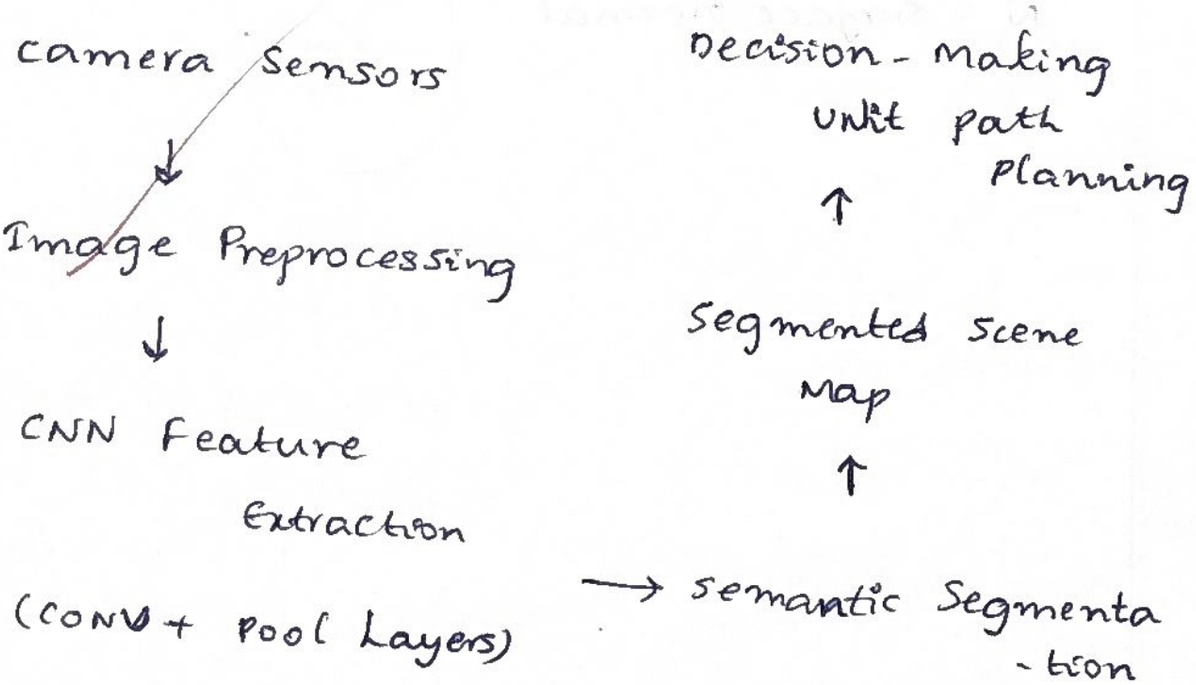
5. Real-Time Interpretation Of Dynamic Environ-ment.

The system continuously updates segmentatio-n results to track moving objects like pedestr-ians and vehicles, helping the vehicle understand complex urban scenes.

6. Support for Autonomus Driving Decisions

The segmented output is sent to the planning + control system, enabling decisions such as lane keeping, obstacle avoidance, braking, and path planning.

Block Diagram



(4) Demonstrate Photometric Modeling

(2M)

+ Photometric modeling describes how light reflects from object surfaces and how this reflection is captured by a camera in an image.

+ It considers factors such as light source direction, surface orientation (normal), & surface reflectance to determine pixel brightness.

+ A common model is the Lambertian model, where image intensity is given by:

$$I = \rho (L \cdot N)$$

where

I = Image Intensity

ρ = Surface reflectance

L = Light direction

N = surface normal.

19/20

Nice